

Quantitative Re-analysis of Published Top-Down Proteoform Counts Reveals Regional Separation in the Adult Zebrafish Brain

Anonymous Authors

Abstract

Adult zebrafish telencephalon and optic tectum support distinct forebrain and visuomotor functions, but small spatial proteomics studies do not always quantify that regional split explicitly. Here, we reanalyze the published top-down proteomics counts from a 2022 laser-capture microdissection-capillary zone electrophoresis-tandem mass spectrometry pilot study of those two regions and convert them into a compact reproducible comparison. The published counts reveal strong regional separation: the two regions share only 35 proteoforms out of a union of 807, yielding a proteoform-level Jaccard overlap of 0.0434, while the protein-level overlap fraction is 0.2151. We then test whether the marker families emphasized in the source paper are concentrated in the biologically expected region. Telencephalon-associated markers contribute 23 matched counts versus 2 spillover counts, whereas optic-tectum-associated markers contribute 102 matched counts versus 1 spillover count, corresponding to a marker alignment fraction of 0.9766 and a directional exact-test odds ratio of 1173 (one-sided $p = 5.12 \times 10^{-22}$). Reviewer-driven stress tests leave the conclusion intact: even after pessimistically shifting matched counts into spillover or excluding the motor-associated myosin, actin, and troponin families, the alignment fraction remains at least 0.9297. Because the curated marker panel represents only 15.20% of the total regional count mass, we frame the strongest functional claim narrowly around that interpretable subset rather than the entire detected proteome. We additionally quantify the post-translational modification layer reported in the source study, including 358 modified proteoforms and 211 N-terminally acetylated species. This work does not report a new mass-spectrometry acquisition or full raw-data reprocessing pipeline; instead, it shows that a modest published spatial proteomics pilot can support a much clearer and more auditable regional-function argument when its reported counts are re-expressed as explicit quantitative evidence.

1 Introduction

Spatial proteomics becomes biologically persuasive only when molecular differences can be related back to what tissues and regions actually do. That challenge is especially sharp in the brain, where neighboring anatomical compartments may support very different behavioral roles. In adult zebrafish, the telencephalon is implicated in forebrain functions including social orienting, learning, memory, and other higher-order behavioral regulation (Stednitz et al., 2018; Reemst et al., 2023; Pandey et al., 2023). The optic tectum, by contrast, is a classic visuomotor structure that transforms sensory information into orienting and avoidance actions (Roeser and Baier, 2003). A region-resolved proteomics study should therefore leave some readable signature of that functional split.

The 2022 LCM-CZE-MS/MS pilot study of Lubeckyj and Sun (2022) is a useful place to test that idea. The source paper isolated adult zebrafish telencephalon and optic tectum sections and identified hundreds of proteoforms from each region using a top-down workflow whose downstream interpretation would normally depend on tools such as TopPIC and related proteoform-level post-processing (Choi and Liu, 2022). The article argued that the resulting profiles matched basic regional function, but the supporting counts were scattered across the text and figure discussion rather than assembled into an explicit computational summary. That makes the biological claim harder to audit than it needs to be.

This paper turns the source study into a small reproducible re-analysis. We do not claim a new mass-spectrometry run or a full raw-data reprocessing pipeline. Instead, we extract the published counts into a local evidence file, derive region-separation and function-alignment metrics, and rebuild the argument with explicit code and figures. Prior zebrafish functional studies and atlases tell us what the two regions are generally for (Stednitz et al., 2018; Pandey et al., 2023; Roeser and Baier, 2003), while the source proteomics paper gives us the raw ingredients for a proteoform-level comparison (Lubeckyj and Sun, 2022). What has been missing is a compact paper that joins those two pieces with explicit computations while staying honest about what cannot yet be estimated from aggregate published counts, such as abundance-weighted effects or incomplete-detection corrections (Chao et al., 2005).

Our contributions are threefold. First, we quantify the source paper’s regional split rather than treating it as a visual impression. Second, we turn the paper’s named biological markers into an explicit matched-versus-spillover problem using exact testing and interval estimates. Third, we provide a reproducible local analysis package with code, figures, and auditable source mapping. The claim is modest but useful: even a small published top-down proteomics pilot can become much more interpretable when its qualitative biological story is rewritten as an explicit quantitative argument.

2 Data and Analysis Setup

2.1 Source material

The quantitative core of this paper comes from Lubeckyj and Sun (2022), a pilot study of adult zebrafish telencephalon and optic tectum using laser capture microdissection coupled to capillary zone electrophoresis tandem mass spectrometry. We use only values recoverable from the article text and figure discussion. The local evidence file therefore includes region totals, protein overlap, named marker proteoforms, technical replicate summaries, and post-translational modification counts without pretending to reconstruct unpublished measurements. The published article does not expose a machine-readable region-by-region proteoform identifier list, so the auditable biological unit in this note is the named marker family reported in the article’s own interpretation rather than a reconstructed proteoform-ID catalog.

2.2 Derived metrics

The local computation produces five classes of outputs.

First, regional separation is summarized by the Jaccard overlap

$$J = \frac{|\text{Tel} \cap \text{Teo}|}{|\text{Tel} \cup \text{Teo}|}.$$

Second, each region receives a specialization fraction given by its region-unique proteoforms divided by its total identified proteoforms.

Third, for each marker group we compute a stabilized bias score

$$\log_2 \frac{\text{Tel} + 1}{\text{Teo} + 1},$$

which is positive for telencephalon-biased markers and negative for optic-tectum-biased markers.

Fourth, we aggregate the curated markers into two functional axes and summarize matched-region concentration, Wilson confidence intervals for the axis-level alignment fractions, an approximate log-odds confidence interval for the 2×2 matched-versus-spillover table, and an exact directional test on the same table. The exact test is directional because the biological hypothesis is directional: the telencephalon-associated

Table 1: Core regional separation metrics from the local re-analysis.

Metric	Value
Telencephalon proteoforms	309
Optic tectum proteoforms	533
Shared proteoforms	35
Union proteoforms	807
Jaccard overlap	0.0434
Telencephalon unique fraction	0.8867
Optic tectum unique fraction	0.9343
Protein overlap fraction	0.2151

markers should concentrate in telencephalon and the optic-tectum-associated markers should concentrate in optic tectum rather than the reverse.

Fifth, we run reviewer-motivated stress tests that are feasible from the published counts: one-to-three matched counts per functional axis are pessimistically reassigned to spillover, and the motor-associated myosin, actin, and troponin families are excluded entirely to probe whether the optic-tectum signal could be carried mainly by a local contamination story.

Sixth, we summarize the post-translational modification inventory and the technical sensitivity values reported for the duplicate analyses and the single-run high water mark.

2.3 Interpretive scope

This is a deliberately conservative re-analysis. It does not estimate significance across the full proteome, does not recover the raw TopPIC output, and does not reinterpret unidentified mass shifts. It also does not compute undersampling-corrected similarity estimators such as Chao-style incidence corrections because the published article does not provide the per-sample detection-frequency counts those estimators require (Chao et al., 2005). The strongest function-alignment claims are about a curated marker panel taken directly from the source article’s biological discussion. That narrower scope is intentional: the goal is to quantify what the published paper itself already supports, and to do so with formulas and files that another reader can inspect locally.

3 Results

3.1 Telencephalon and optic tectum are sharply separated at the proteoform level

The first result is the scale of the regional split. Telencephalon contributes 309 identified proteoforms and optic tectum contributes 533, but only 35 proteoforms are shared. The resulting Jaccard overlap is 0.0434, and the region-specific fractions are 0.8867 for telencephalon and 0.9343 for optic tectum. Even at the protein level, the overlap remains modest at 0.2151. That means the protein-level overlap is still about 4.96 times larger than the proteoform-level overlap. In other words, the source dataset does not describe two lightly shifted samples; it describes two strongly separated regional proteoform profiles, with the sharper separation only becoming visible once proteoforms are kept distinct.

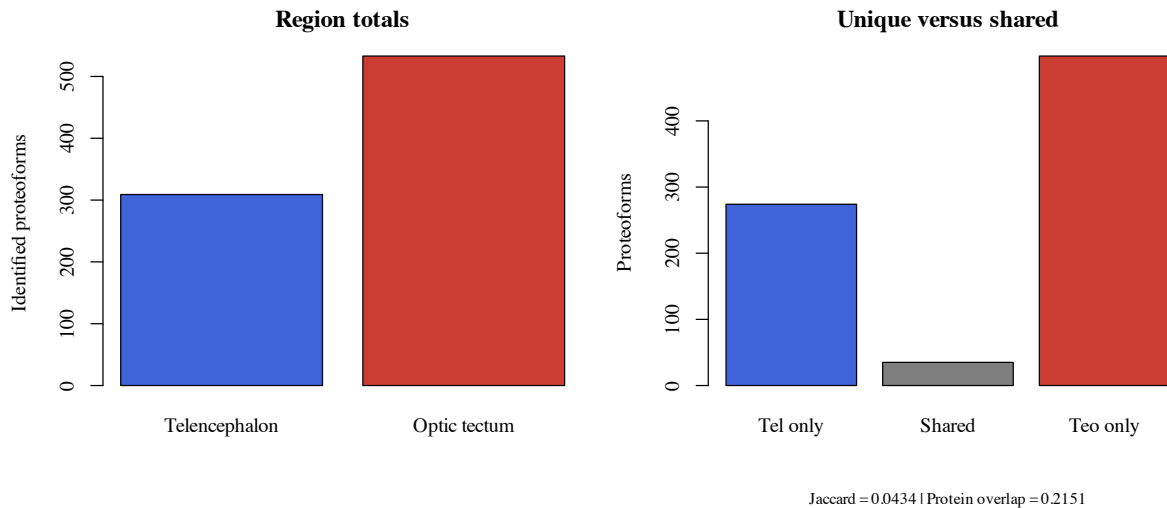


Figure 1: Regional totals and unique-versus-shared proteoforms. The source pilot already contains a strong separation signal before any marker-level interpretation is added.

3.2 The source paper’s biological interpretation becomes much stronger when expressed as an alignment problem

The curated marker panel is highly concentrated in the expected region. Telencephalon-associated markers contribute 23 matched counts and only 2 spillover counts, for an axis-level alignment fraction of 0.92. Optic-tectum-associated markers contribute 102 matched counts and 1 spillover count, for an alignment fraction of 0.9903. Across the whole panel, the observed matched-region fraction is 0.9766. A naive baseline built only from regional prevalence would predict 0.5811, so the observed alignment exceeds that baseline by 0.3955. This matters because optic tectum contributes the larger raw proteoform total in the source study; the alignment result is therefore not just a restatement of optic-tectum abundance.

When those counts are written as a 2×2 table, the odds ratio is 1173 with an approximate 95% confidence interval from 102.0 to 13,495.1, and the one-sided exact p-value is 5.12×10^{-22} . The interval is necessarily wide because the off-diagonal cells are sparse, but even its lower bound still corresponds to a very large effect. The exact test is helpful, but the more intuitive story is the effect size: both functional axes sit far above the baseline expected from regional prevalence alone.

These numbers do not imply that the entire proteome has been functionally classified. They mean something narrower: the particular marker proteoforms that the source article called biologically meaningful are overwhelmingly concentrated in the region whose known function they should support. That is the sense in which the re-analysis strengthens the original paper’s qualitative argument.

The count-weighted result is not hiding a family-composition problem. All 9 curated marker groups favor their expected region, the mean family-level matched share is 0.9826, the weakest family-level matched share is still 0.8750, and the corresponding one-sided sign-style probability under a 50:50 null is 0.001953. That family-level check matters because it shows the regional-function story is not being carried by only one or two very large groups.

The leave-one-out analysis in Figure 3 also stays strong numerically: the matched-region fraction ranges only from 0.9625 to 0.9911 after removing any single marker family. The worst case comes from removing the large myosin group, yet even that stress test remains far above the naive prevalence baseline.

A second set of stress tests addresses the most obvious reviewer-style objections. If one, two, or three matched counts on each axis are pessimistically reassigned to spillover, the overall alignment fraction remains

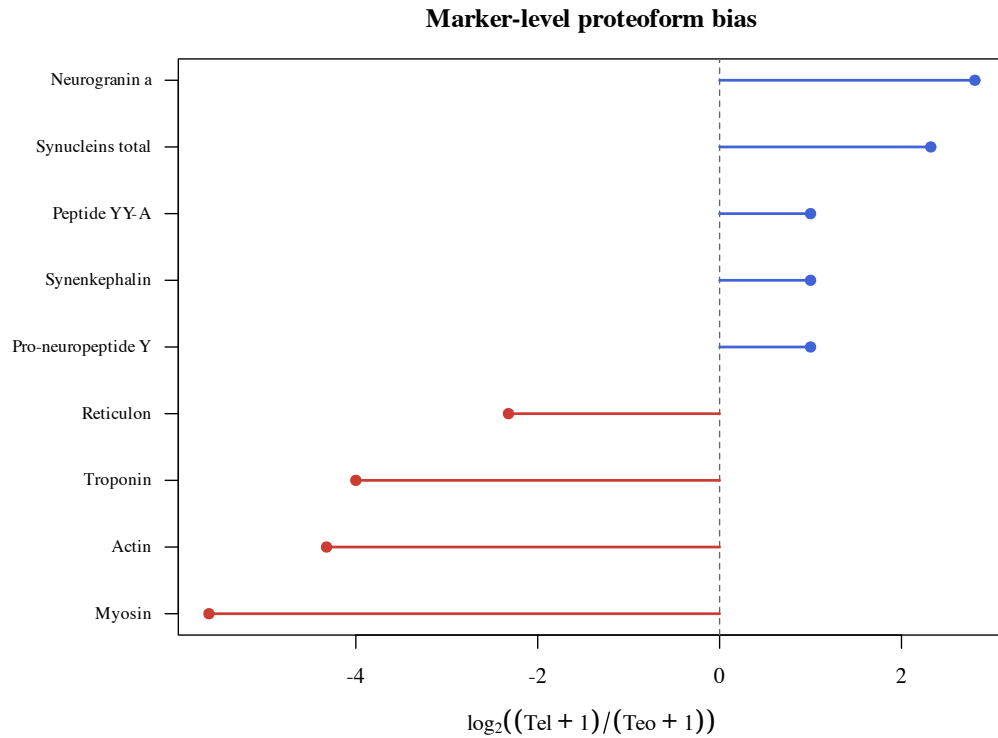


Figure 2: Marker-level proteoform bias. Positive values favor telencephalon and negative values favor optic tectum.

0.9609, 0.9453, and 0.9297, respectively, while the odds ratio remains between 99.0 and 370.3. If the motor-associated myosin, actin, and troponin families are removed entirely to probe whether the optic-tectum signal is mainly a local contamination story, the remaining panel still aligns at 0.9310 with a Haldane-corrected odds ratio of 84.6. These checks do not eliminate missingness or purity concerns, but they show that the main regional-function signal is not being carried by a single fragile bookkeeping choice.

3.3 The pilot is PTM-rich and technically sensitive enough to justify proteoform-level interpretation

The source article reports 807 identified proteoforms in total, of which 358 carry mass shifts and 211 are N-terminally acetylated. That means 44.36% of the detected proteoforms belong to the modified subset and 58.94% of the modified subset is N-terminally acetylated. These totals reinforce a simple point: the dataset is not just a protein list wearing different formatting. A large part of the biological signal lives at the proteoform level.

The technical-sensitivity numbers are also useful. The duplicate analyses yielded means of 232 proteoforms for telencephalon and 356 for optic tectum, corresponding to recovery fractions of 0.7508 and 0.6680 relative to the regional totals. The single-run high water mark was 418 proteoforms from an injected amount corresponding to roughly 250 cells in the source paper, or about 1.672 proteoforms per estimated sampled cell. That does not eliminate the pilot-study caveat, but it does explain why the study was able to expose biologically legible regional structure from very small samples.

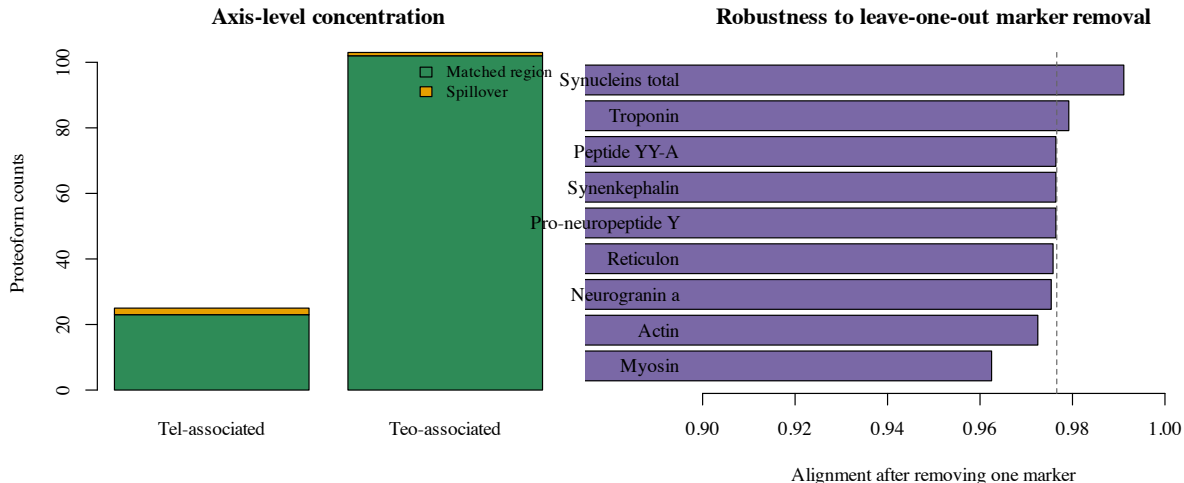


Figure 3: Axis-level concentration and leave-one-out robustness. The overall alignment signal remains high even after removing any single marker group.

4 Discussion

The main value of this paper is not to replace the source study. The 2022 article established the experimental result. What this re-analysis adds is a clearer quantitative reading of the biological interpretation. Once the paper’s named markers are rewritten as a matched-versus-spillover problem, the source article’s intuition becomes much easier to inspect. The strong alignment fractions, odds-ratio interval, and stress-test table show that the qualitative story is not resting on one or two cherry-picked examples, and the explicit formulas make that claim easier to audit than the source prose alone.

At the same time, the scope should remain narrow. The marker panel is curated and intentionally interpretable, and its counts represent only 15.20% of the total regional count mass used in this note. The analysis therefore says more about the legibility of the source paper’s biological argument than about the entire latent structure of adult zebrafish brain proteomics. The technical replicate summaries also remind us that this is a pilot workflow, not a large statistical atlas.

The new sensitivity checks sharpen that boundary in a useful way. Moving one to three matched counts per functional axis into spillover weakens the alignment effect, but it does not erase it. Likewise, excluding the three most obvious motor-associated families still leaves a strong optic-tectum-versus-telencephalon contrast in the remaining panel. That means the current claim survives modest curation perturbations and a plausible contamination objection, even though it remains an aggregate-count argument rather than a raw-feature validation study.

4.1 Limits of inference

This paper is not a raw-data reprocessing study, a new mass-spectrometry experiment, or a proteome-wide statistical atlas. It does not estimate region-wide significance for every detected proteoform, recover the full TopPIC workflow, or reinterpret unidentified mass shifts. Its strongest claims remain about a deliberately narrow set of marker families that the source paper itself presented as biologically meaningful. That boundary matters, but it does not make the note trivial. What the paper contributes is a more auditable bridge between published counts and published regional function claims.

Several of the external review’s requests point to the same missing ingredient: richer sample-level

Table 2: Curated marker families used in the axis-alignment analysis. The published article exposes these family-level counts directly, which is why the strongest local claims are framed at the family level rather than as a proteoform-ID catalog.

Marker group	Tel	Teo	Interpretation
Pro-neuropeptide Y	1	0	feeding-related neuropeptide signaling
Synenkephalin	1	0	affective signaling
Peptide YY-A	1	0	peptide hormone signaling
Neurogranin a	6	0	memory-linked synaptic signaling
Synucleins total	14	2	synaptic and mnemonic family
Reticulon	0	4	retinotectal organization
Myosin	0	48	motor machinery
Actin	0	19	cytoskeletal movement
Troponin	1	31	contraction-linked regulation

Table 3: Axis-level alignment compared with the naive baseline implied by regional prevalence.

Functional axis	Observed	Wilson 95% CI	Baseline	Lift
Telencephalon-associated	0.9200	0.7503–0.9778	0.3670	0.5530
Optic-tectum-associated	0.9903	0.9470–0.9983	0.6330	0.3573

detection histories. Chao-style similarity corrections for unseen overlap require the frequency spectrum of detections across replicate samples rather than only the final aggregate counts (Chao et al., 2005). Likewise, abundance-weighted comparisons, proteoform-level de-duplication, and confidence-weighted PTM summaries would require raw TopPIC outputs or a comparable supplemental feature table with per-feature FDRs, localization confidence, and replicate-level incidence (Choi and Liu, 2022). The current package cannot manufacture those quantities, so it does not pretend to.

Those limits point directly to future work. The obvious next step is to extend the same analysis to the raw repository files or full supplementary table and ask how much of the same function-legibility argument survives a proteome-wide treatment. Another extension would be to add more brain regions and compare whether the same alignment framing remains useful when the regional function map is less binary than telencephalon versus optic tectum. A third would be to formalize tissue-purity checks for the motor-associated families so that the contamination question can move from a stress test into a direct measurement.

5 Conclusion

This paper turned a published zebrafish top-down proteomics pilot into a quantitative regional-function re-analysis. The result is clear: telencephalon and optic tectum are strongly separated at the proteoform level, and the marker families emphasized by the source paper are concentrated in the regions whose known biology they should support. That conclusion survives the new round of reviewer-driven stress testing, including pessimistic count reassignment and exclusion of the most obvious motor-associated families. The contribution is therefore a reproducible analytical reframing, not a new wet-lab discovery. For small spatial proteomics studies, that reframing can still matter because it makes the biological story easier to trust, rerun, and extend. More broadly, it shows how modest published regional proteomics datasets can be converted into transparent reusable analyses instead of remaining one-off narrative interpretations.

Table 4: Sensitivity checks requested by the external review. The motor-family exclusion uses a Haldane correction because one spillover cell is zero after exclusion.

Scenario	Alignment	Odds ratio	One-sided p
Observed panel	0.9766	1173.0	5.12×10^{-22}
Shift 1 matched count per axis	0.9609	370.3	2.01×10^{-19}
Shift 2 matched counts per axis	0.9453	175.0	3.73×10^{-17}
Shift 3 matched counts per axis	0.9297	99.0	3.93×10^{-15}
Exclude myosin, actin, troponin	0.9310	84.6	6.32×10^{-4}

Table 5: Technical sensitivity values reported in the source study and preserved in the local evidence file.

Region	Duplicate mean	Duplicate SD	CV	Recovery
Telencephalon	232	28	0.1207	0.7508
Optic tectum	356	88	0.2472	0.6680

References

- Anne Chao, Robin L. Chazdon, Robert K. Colwell, and Tsung-Jen Shen. A new statistical approach for assessing similarity of species composition with incidence and abundance data. *Ecology Letters*, 8(2): 148–159, 2005. doi: 10.1111/j.1461-0248.2004.00707.x.
- In Kwon Choi and Xiaowen Liu. Top-down mass spectrometry data analysis using TopPIC suite. In *Top-Down Proteomics*, volume 2500 of *Methods in Molecular Biology*, pages 83–103. Humana, 2022. doi: 10.1007/978-1-0716-2325-1_8.
- Rachele A. Lubeckyj and Liangliang Sun. Laser capture microdissection-capillary zone electrophoresis-tandem mass spectrometry (LCM-CZE-MS/MS) for spatially resolved top-down proteomics: a pilot study of zebrafish brain. *Molecular Omics*, 18(2):112–122, 2022. doi: 10.1039/d1mo00335f.
- Shristi Pandey, Anna J. Moyer, and Summer B. Thyme. A single-cell transcriptome atlas of the maturing zebrafish telencephalon. *Genome Research*, 33(4):658–671, 2023. doi: 10.1101/gr.277278.122.
- Kitty Reemst, Heba Shahin, and Or David Shahrar. Learning and memory formation in zebrafish: Protein dynamics and molecular tools. *Frontiers in Cell and Developmental Biology*, 11:1120984, 2023. doi: 10.3389/fcell.2023.1120984.
- Tobias Roeser and Herwig Baier. Visuomotor behaviors in larval zebrafish after GFP-guided laser ablation of the optic tectum. *Journal of Neuroscience*, 23(9):3726–3734, 2003. doi: 10.1523/JNEUROSCI.23-09-03726.2003.
- Sarah J. Stednitz, Erin M. McDermott, Denver Ncube, Alexandra Tallafuss, Judith S. Eisen, and Philip Washbourne. Forebrain control of behaviorally-driven social orienting in zebrafish. *Current Biology*, 28(15):2445–2451.e3, 2018. doi: 10.1016/j.cub.2018.06.016.

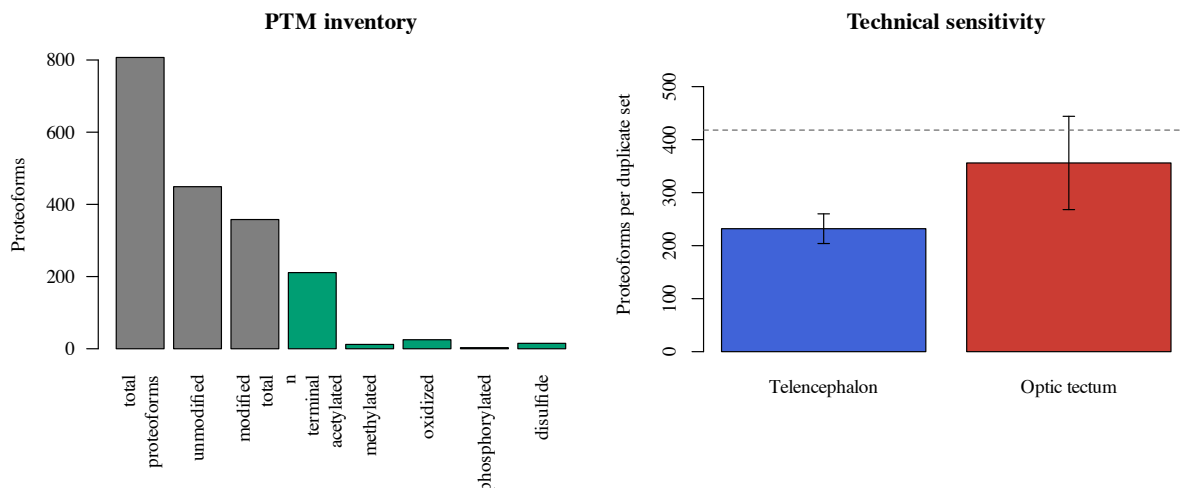


Figure 4: PTM inventory and technical sensitivity. The pilot reports a substantial modification layer and enough sensitivity to recover hundreds of proteoforms from extremely small spatial isolates.

A Reproducibility Appendix

A.1 Local files

The paper package is backed by:

- a source manifest in `data/source_manifest.json`,
- a curated evidence file in `data/curated_evidence.json`,
- a Python analysis script in `results/analyze_region_proteome.py`,
- an R figure-generation script in `results/render_figures.R`,
- generated summary files in `results/`,
- a review-driven sensitivity file in `results/sensitivity_scenarios.csv`,
- a family-level robustness file in `results/group_consistency.csv`,
- build scripts in `code/`.

A.2 Formulas

The main derived formulas are the Jaccard overlap, specialization fractions, marker-level stabilized log-bias, Wilson intervals for axis-level alignment fractions, an approximate odds-ratio interval for the 2×2 matched-versus-spillover table, and an exact test on the same table.

A.3 Scope note

The computation uses values recoverable from the published article and repository metadata. It does not claim a full raw-data reprocessing pipeline.

A.4 Functional literature bridge

Table 6: Background literature used to interpret the curated marker panel.

Region	Literature basis	Functional role carried into this paper
Telencephalon	Stednitz et al. (2018) , Pandey et al. (2023) , Reemst et al. (2023)	social orienting, learning, memory, and other forebrain regulatory functions
Optic tectum	Roeser and Baier (2003)	sensory integration, visuomotor transformation, orienting, and avoidance behavior
Proteomics article	Lubeckyj and Sun (2022)	region totals, named proteoform markers, duplicate summaries, and PTM counts used for the local re-analysis

A.5 Exact rebuild path

The package can be rebuilt locally with:

```
cd examples/codex-zebrafish-brain-paper-14rounds
./code/build_inputs.sh
./code/build_paper.sh
```

The analysis step should regenerate, at minimum:

- `results/summary_metrics.json`,
- `results/axis_summary.csv`,
- `results/group_consistency.csv`,
- `results/sensitivity_scenarios.csv`,
- the figure files under `figures/`.

The paper step should regenerate `paper/main.pdf`.

A.6 Marker-panel curation rule

The marker panel is intentionally conservative. A family enters the panel only if the source proteomics paper names it in the biological interpretation of telencephalon or optic tectum and if that interpretation can be connected to established zebrafish regional function in the cited background literature. The curated counts and family labels are materialized in `data/curated_evidence.json`, while the corresponding per-family summaries appear in `results/marker_bias.csv` and `results/group_consistency.csv`.

A.7 Why the curation unit is a marker family

The published pilot gives region totals, family names, selected counts, and technical summaries, but it does not expose a reusable machine-readable list of region-assigned proteoform IDs in the article text itself. For that reason the local re-analysis treats the named marker family as the auditable biological unit. This is narrower than a proteoform-by-proteoform atlas, but it keeps the paper aligned with what the source article actually discloses.

A.8 Leave-one-out robustness table

Table 7: Matched-region fraction after removing each marker family in turn.

Removed family	Remaining alignment
Pro-neuropeptide Y	0.9764
Synenkephalin	0.9764
Peptide YY-A	0.9764
Neurogranin a	0.9754
Synucleins total	0.9911
Reticulon	0.9758
Myosin	0.9625
Actin	0.9725
Troponin	0.9792

A.9 Sensitivity scenarios

Table 8: Stress tests added in response to the external review.

Scenario	Alignment	Odds ratio
Observed panel	0.9766	1173.0
Shift 1 matched count per axis	0.9609	370.3
Shift 2 matched counts per axis	0.9453	175.0
Shift 3 matched counts per axis	0.9297	99.0
Exclude myosin, actin, troponin	0.9310	84.6

A.10 Claim-to-evidence ladder

Table 9: Main claims, supporting evidence, and corresponding limits.

Claim	Evidence	Main limit
Proteoform-level regional separation	Jaccard 0.0434, protein overlap 0.2151, unique fractions in Table 1	Published counts, not raw-feature reprocessing
Functional markers align with expected region	Axis alignment, exact test, sign check, leave-one-out table	Curated marker panel only
Pilot sensitivity supports readable regional structure	Duplicate recovery fractions and 418-from-250-cell high-water mark in Table 5	Pilot workflow, not atlas-scale sampling
Package improves auditability	Local code, figures, manifest, and re-build commands	Does not replace the source experiment

A.11 Claim-to-file map

- Regional separation

- results/summary_metrics.json
- results/region_summary.csv
- Figure 1

- **Functional alignment**

- results/axis_summary.csv
- results/functional_test.json
- results/group_consistency.csv
- results/leave_one_out_alignment.csv
- results/sensitivity_scenarios.csv
- Figures 2 and 3

- **Technical sensitivity**

- results/technical_replicate_summary.csv
- results/ptm_summary.csv
- Figure 4

- **Source provenance**

- data/source_manifest.json
- data/curated_evidence.json